Differentiation of Implicit Functions
(隱函數微分)

若一個函數難以用 y (或 x) 為主項，則它就被稱為隱函數 (implicit functions)，

例如：

(1) $x^2+y^2-3x+5=0$ (2) $x^2-xy+y=1$

我們可以將隱函數的每一項加以微分，再以主項變換而求得 $\frac{dy}{dx}$ 。

e.g.1 If $x^2+y^2-3x+5=0$, find $\frac{dy}{dx}$.

$$x^2+y^2-3x+5=0$$

$$\frac{d}{dx}(x^2+y^2-3x+5)=\frac{d}{dx}(0)$$

$$2x+2y\frac{dy}{dx}-3=0$$

$$2y\frac{dy}{dx}=3-2x$$

$$\frac{dy}{dx}=\frac{3-2x}{2y}$$

Important Note!!

$$\frac{d}{dx}(y^2)=\frac{dy^2}{dx}$$

$$=\frac{dy^2}{dy}\cdot\frac{dy}{dx}$$

$$=2y\frac{dy}{dx}$$

e.g.2 If $x^2-xy+y=1$, find $\frac{dy}{dx}$.

$$x^2-xy+y=1$$

$$\frac{d}{dx}(x^2-xy+y)=\frac{d}{dx}(1)$$

$$2x-(y+x\frac{dy}{dx})+\frac{dy}{dx}=0$$

$$(1-x)\frac{dy}{dx}=y-2x$$

$$\frac{dy}{dx}=\frac{y-2x}{1-x}$$

Important Note 2!!

$$\frac{d}{dx}(xy)$$

$$=y\frac{d}{dx}(x)+x\frac{d}{dx}(y)$$

$$=y+x\frac{dy}{dx}$$

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Differentiation of Implicit Functions

隱函數之微分變化多端，以下有更多例子：

	Implicit Function	Differentiate with respect to x
e.g.1.	y^3	$3y^2\frac{dy}{dx}$
e.g.2.	y^n (n is constant)	$ny^{n-1}\frac{dy}{dx}$
e.g.3.	xy	$y+x\frac{dy}{dx}$
e.g.4.	x^2y^3	$2xy^3+3x^2y^2\frac{dy}{dx}$
e.g.5.	x^my^n (m, n are constant)	$mx^{m-1}y^n+nx^my^{n-1}\frac{dy}{dx}$
e.g.6.	$\frac{x}{y}$	$\frac{y-x\frac{dy}{dx}}{y^2}$
e.g.7.	$\sin y$	$\cos y\frac{dy}{dx}$
e.g.8.	$\cos^2 y$	$-2\cos y\sin y\frac{dy}{dx}$
e.g.9.	$\tan(xy)$	$\sec^2(xy)(y+x\frac{dy}{dx})$
e.g.10.	e^y	$e^y\frac{dy}{dx}$
e.g.11.	e^{xy}	$e^{xy}(y+x\frac{dy}{dx})$
e.g.12.	$e^{\frac{x}{y}}$	$e^{\frac{x}{y}}\left(\frac{y-x\frac{dy}{dx}}{y^2}\right)$
e.g.13.	$\ln y$	$\frac{1}{y}\frac{dy}{dx}$
e.g.14.	$\ln(xy)$	$\frac{1}{xy}(y+x\frac{dy}{dx})$

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HKCEE Questions (Level 2)

1. ***[A.Math 2005 Q.9(b)]***

Let $xy + y^2 = 2005$. Find $\frac{dy}{dx}$. ***(3 marks)***

2. ***[A.Math 2003 Q.4]***

Given that $3x^2 + 3y^2 - 2xy = 12$. Find $\frac{dy}{dx}$ when $x = 2$, $y = 0$. ***(4 marks)***

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HKCEE Questions (Level 2)

3. ***[A.Math 1997 Paper I Q.2]***

$P(8, 1)$ is a point on the curve $y^2 + \sqrt[3]{x}y - 3 = 0$. Find the value of $\frac{dy}{dx}$ at P.

(3 marks)

4. ***[A.Math 2002 Q.3]***

Let $x\sin y = 2002$. Find $\frac{dy}{dx}$. ***(3 marks)***

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HKDSE Questions (level 2)

5.. ***[M2 2014 Q.4 Modified]***

Let $x = 2y + \sin y$. Find $\dfrac{dy}{dx}$ in terms of y. ***(3 marks)***

6. ***[M&S 2000 Q.1]***

Let $\ln(xy) = \dfrac{x}{y}$ where $x, y > 0$. Show that $\dfrac{dy}{dx} = \dfrac{xy - y^2}{xy + x^2}$ ***(4 marks)***

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HKCEE Questions (Level 2)

7. ***[A.Math 1993 Paper I Q.7]***

Given the curve C: $x^2 - 2xy^2 + y^3 + 1 = 0$.

(a) Find $\frac{dy}{dx}$

(b) Find the equation of the tangent to C at the point (2, –1). ***(7 marks)***

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HKCEE Questions (Level 2)

8. ***[A.Math 2009 Q.8]***

C is a curve with equation $y^3 + x^3 y = 0$.

(a) Find $\frac{dy}{dx}$.

(b) Find the equation of the tangent to the curve C at the point (1, 2).

(5 marks)

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HKCEE Questions (level 2)

9. ***[A.Math 2001 Q.7]***

$P(2, 0)$ is a point on the curve $x-(1+\sin y)^5=1$.

Find the equation of the tangent to the curve at point P. ***(5 marks)***

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HKCEE Questions (level 2)

10. ***[A.Math 1994 Paper I Q.6]***

Given the curve C: $x^2 + y\cos x - y^2 = 0$.

(a) Find $\frac{dy}{dx}$.

(b) $P(\frac{\pi}{2}, -\frac{\pi}{2})$ is a point on the curve C. Find the equation of the tangent to the curve at P. ***(7 marks)***

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HKDSE Questions (level 2)

11. ***[M2 2014 Q.3]***

Find the equation of the tangent to the curve $x\ln y + y = 2$ at the point where the curve cuts the *y*-axis. ***(5 marks)***

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HKCEE Questions (Level 2)

12. ***[A.Math 1990 Paper I Q.7]***

Given the curve C: $x^2 + 4xy + 5y^2 = 1$. Find $\frac{dy}{dx}$.

Hence find equations of the two tangents to C which are parallel to the line $y = -\frac{1}{2}x$.

(7 marks)

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HKCEE Questions (Level 2)

13. ***[A.Math 2006 Q.12]***

(a) Let $x^2 - xy + y^2 = 7$. Find $\frac{dy}{dx}$.

(b) Find the equation of the normal to the curve $x^2 - xy + y^2 = 7$ at the point (1, 3). ***(5 marks)***

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HKCEE Questions (Level 2)

14. ***[A.Math 1995 Paper I Q.6]***

$P(4, 1)$ is a point on the curve $y^2 + y\sqrt{x} = 3$, where $x > 0$.

(a) Find the value of $\frac{dy}{dx}$ at P.

(b) Find the equation of the normal to the curve at P. ***(7 marks)***

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HKASL Questions (Level 2)

15. ***[M&S 1999 Q.1]***

It is given that $e^{xy} = \dfrac{x(x+1)^3}{x^2+1}$, where $x > 0$.

(a) Find the value of y when $x = 1$.

(b) Find the value of $\dfrac{dy}{dx}$ when $x = 1$. ***(5 marks)***

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HKASL Questions (level 2)

16. ***[M&S 2008 Q.2]***

Suppose $y^3 - uy = 1$ and $u = 2^{x^2}$.

(a) Find $\frac{dy}{du}$ in terms of u and y.

(b) Find $\frac{du}{dx}$ in terms of x.

(c) Find $\frac{dy}{dx}$ in terms of x and y. ***(7 marks)***

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More Questions in Past Papers

1. ***[A.Math 1986 Paper I Q.4]***

Find the equation of the tangent to the curve $x^2 + xy + y^2 = 7$ at the point (2, 1).

(6 marks)

2. ***[A.Math 1988 Paper I Q.2]***

Find the equations of the two tangents to the curve $y^2 = x^2 y + 2$ at the points where $x = 1$. ***(5 marks)***

3. ***[A.Math 2000 Paper I Q.4]***

$P(-1, 2)$ is a point on the curve $(x+2)(y+3) = 5$.

(a) Find the value of $\frac{dy}{dx}$ at P.

(b) Find the equation of the tangent to the curve at P ***(5 marks)***

4. ***[A.Math 1992 Paper I Q.5]***

The curve $(x-2)(y^2+3) = -8$ cuts the y-axis at two points.

(a) Find the coordinates of the two points;

(b) Find the slope of the tangent to the curve at each of the two points.

(6 marks)

5. ***[A.Math 1999 Paper I Q.6]***

The point $P(a, a)$ is on the curve $3x^2 - xy - y^2 - a^2 = 0$, where a is a constant.

(a) Find the value of $\frac{dy}{dx}$ at P.

(b) Find the equation of the tangent to the curve at P ***(6 marks)***

6. ***[A.Math 1998 Q.8]***

$P(0, 2)$ is a point on the curve $x^2 - xy + 3y^2 = 12$

(a) Find the value of $\frac{dy}{dx}$ at P.

(b) Find the equation of the normal to the curve at P. ***(7 marks)***